Who benefits most?

Sorted effects, step by step · A colon cancer trial

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Session overview

1. Is the effect the same for everybody? Today's papers.
2. Sorted effects, step by step, with the R code: a colon cancer trial.
3. Can we trust the pattern? Uniform bands, model choice and held-out patients.
4. Three trials side by side, and discussion.

Is the effect the same for everybody?

01

A trial reports one average effect. Whom does it describe?

One average, many patients

- A trial answers: does the treatment work **on average**?
- Patients differ, so the effect can differ too: large for some, small or none for others.
- So we also want to know: **who benefits most, and who least?**

“Why not … just estimate the distribution of effects?” *The Effect*, §22.3 (Huntington-Klein 2025)

Today's papers

Method · Econometrica 2018
The sorted effects method: discovering heterogeneous effects beyond their averages
Chernozhukov, Fernández-Val, and Luo (2018)

- Sorted partial effects: percentiles of the effect across patient profiles
- Classification analysis: who is most and least affected
- Bootstrap bands for the whole curve

Software · The R Journal 2020
SortedEffects: sorted causal effects in R
Chen et al. (2020)

- `spe()`: sorted effects with uniform bands
- `ca()`: compare the most and least affected
- Worked example: mortgage denials (backup)

Intuition · The Effect 2025, §22.3
The Effect: an introduction to research design and causality
Huntington-Klein (2025)

- Model the effect, predict it for everyone, sort, describe
- Free online textbook
- Also causal forests: next time

Same average, different stories

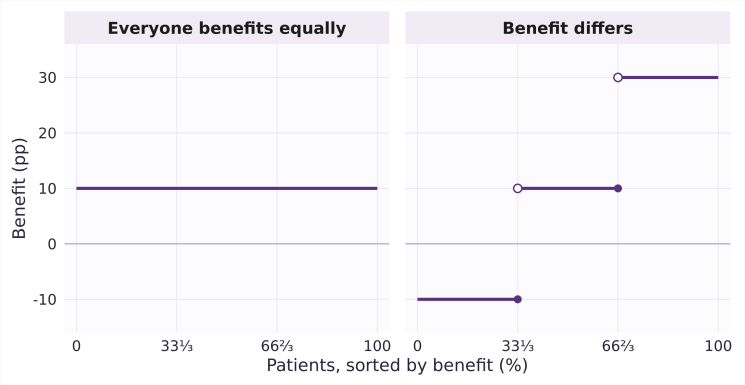
Three equally common kinds of patients, A, B and C. Without treatment, 40% of each die.

Benefit (pp)	A	B	C
Everyone benefits equally	+10	+10	+10
Benefit differs	-10	+10	+30

Both stories: average benefit **10 pp**. The average cannot tell them apart.

Constructed example with known risks, not data. Benefit = risk without treatment − risk with treatment.

Sorting tells them apart

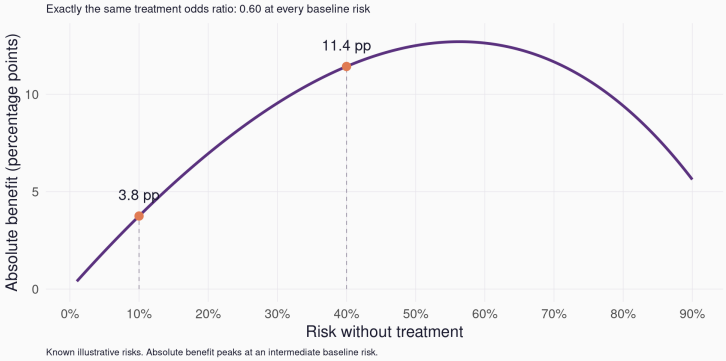


- Line patients up from least to most benefit.
- Everyone benefits equally: a flat line at **10 pp**.
- Benefit differs: a staircase from **−10 to +30 pp**.
- One kind of patient in three (A) has a higher risk with treatment, despite the positive average.

Sorted effects estimate this curve for kinds of patients, not individuals.

Exact quantiles of the constructed example; no estimation.

Even a constant odds ratio gives different benefits



Algebraic illustration with known risks.

- Same odds ratio (**0.60**) for every patient.
- **10%** baseline risk → **3.8 pp**.
- **40%** baseline risk → **11.4 pp**.
- A logistic model already lets benefits differ; interactions let them differ more.

Sorted effects, step by step

02

A colon cancer trial: who benefits from chemotherapy?

The trial

- 619 patients with resected **stage III colon cancer**, randomized to observation or **levamisole + 5-FU**.
- Outcome: **death within three years**.
- Died: **34.7%** with observation (109/314), **25.7%** with chemotherapy (78/304).
- On average, about **9 fewer deaths per 100** treated patients. The same for everybody?

`survival::colon`, from the trial reports (Moertel et al. 1990, 1995); levamisole + 5-FU is a historical regimen. Two of three arms (619 patients); three-year status known for 618; models use the 593 with complete baseline data.

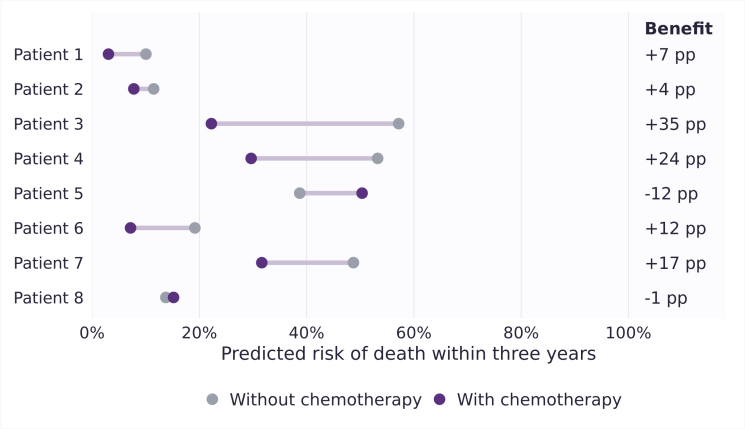
Step 1 · Let the effect differ

- A logistic model for death within three years.
- Chemotherapy's effect may change with **age, sex, number of positive nodes, tumour grade and tumour extent**.
- Bowel obstruction, tumour adherence to nearby organs and time from surgery to registration adjust the baseline risk.

Randomization makes the comparison fair; the model only describes how the effect varies.

Six treatment interactions, fixed before fitting; 19 parameters. Model code in backup.

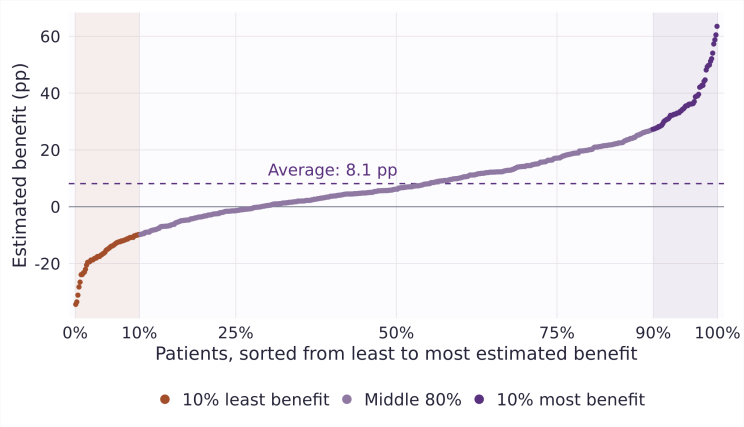
Step 2 · Predict both futures for every patient



- For each patient: predicted risk **without** and **with** chemotherapy.
- The gap is the **estimated benefit**.
- It is an average for patients like them, not a personal effect.

Eight of 593 patients, chosen to span the range and shown unsorted. Point estimates from the Step 1 model.

Step 3 · Sort them



- All 593 patients, from least to most estimated benefit.
- Average: **8.1 pp**. Middle 80%: **−10 to +27 pp**.
- Only **15%** have an estimated benefit within 3 pp of the average.
- The extremes are the least certain estimates.

Point estimates; uncertainty follows in chapter 03.

In symbols: three estimands

$$\Delta(x) = \mathsf{Pr}(Y = 1 \mid A = 0, X = x) - \mathsf{Pr}(Y = 1 \mid A = 1, X = x)$$

$$\mathsf{APE} = E[\Delta(X)], \qquad \mathsf{SPE}(u) = Q_{\Delta(X)}(u), \quad u \in [0.1,\, 0.9]$$

$$\mathsf{CA}(Z) = E[Z \mid \Delta(X) \geq \mathsf{SPE}(0.9)] - E[Z \mid \Delta(X) \leq \mathsf{SPE}(0.1)]$$

- Y: death within three years; A: chemotherapy (1) or observation (0); X: the eight traits in the model.
- Estimates plug the fitted model into Δ for all 593 patients; CA compares a trait Z between the two 10% tails.
- With randomization, $\Delta(x) = E[Y(0) - Y(1) \mid X = x]$: an average for profile x, not an individual effect.

Notation after Chernozhukov, Fernández-Val, and Luo (2018). SPE is a quantile of effects, not a difference between outcome quantiles.

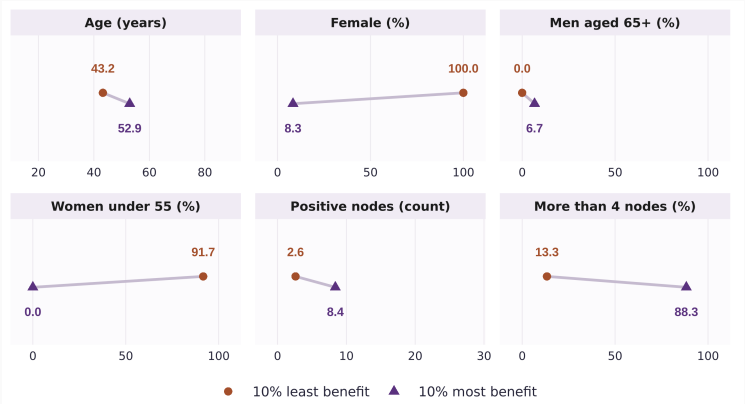
Step 3 in R: spe()

```r
spe_result <- SortedEffects::spe(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  us = seq(.10, .90, .01),
  b = 499,
  alpha = .05, bc = FALSE,
  seed = 420
)
```

Fit the outcome model
Our chosen formula and a logistic link.
Set the contrast
This ordering gives observation − chemotherapy.
Choose the percentiles
10th–90th, in one-percentile steps.
Quantify uncertainty
499 resamples; 95% band; no bias correction.

`fm` is the Step 1 model (code in backup). The band covers the chosen percentile grid; coverage needs valid effect estimates and regularity conditions.

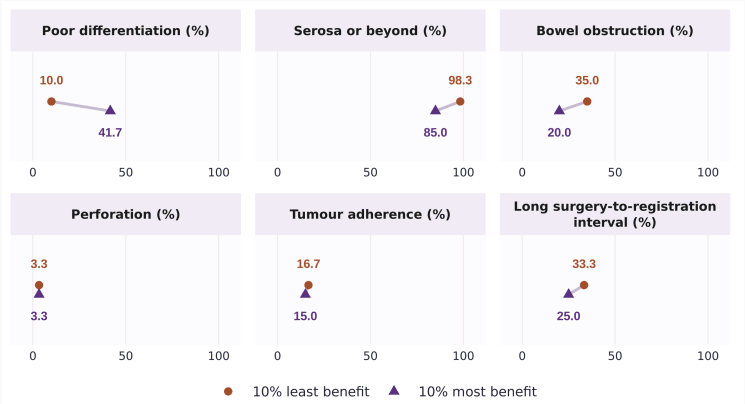
Step 4 · Who does the model rank at each end?



- Least estimated benefit: **all women**, mean age **43**.
- Most estimated benefit: **92% men**, mean age **53**.
- More than four positive nodes: **13% vs 88%**.
- Partly by design: the model lets the effect vary with sex, age and nodes.

The 60 patients (10%) at each end. Points are tail means; the lines are not confidence intervals.

Step 4 · Tumour characteristics



Same 60 + 60 patients. “Adherence”: tumour adherence to nearby organs.

- Poor differentiation: **10% vs 42%**.
- Bowel obstruction: **35% vs 20%**.
- Perforation: **3%** in both tails.

Step 4 in R: ca()

```r
ca_result <- SortedEffects::ca(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  u = .10,
  interest = "moment", cl = "diff",
  t = colon_characteristics()$variable,
  b = 499, alpha = .05,
  bc = FALSE, seed = 420
)
```

Model and contrast
Same mortality model; benefit = observation − chemotherapy.
Choose the tails
Bottom and top 10%: 60 patients each here.
Compare characteristics
High − low means, across all twelve traits.
Bootstrap inference
499 resamples; SEs and p-values adjusted across the twelve traits.

`ca()` is classification analysis. For these data the package's own bootstrap broke down; the p-values on the next slide come from our checked re-run with stable starting values (backup).

Step 4 · How different are the two tails?

Characteristic	10% least benefit	10% most benefit	Difference (SE)	Adjusted p
Women	100%	8%	−92 (21)	.04
Women under 55	92%	0%	−92 (21)	.04
Mean age (years)	43	53	+9.7 (7.1)	.87
Mean positive nodes	2.6	8.4	+5.8 (2.2)	.30
More than 4 positive nodes	13%	88%	+75 (22)	.13
Poor differentiation	10%	42%	+32 (28)	.95

The other six traits: adjusted p ≥ .76.

These tests compare the model's tails. They do not test whether sex modifies the effect of chemotherapy.

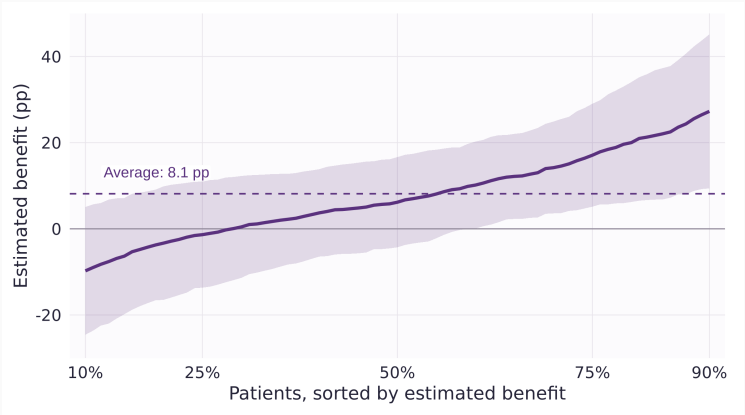
Differences of percentages in pp. p-values adjusted jointly across all 12 traits (499 bootstrap resamples). The package's own bootstrap broke down for these data, so they come from our checked re-run with stable starting values (backup).

Can we trust the pattern?

03

Is the spread real, or noise from one model and one trial?

How sure are we?



- Shaded: a 95% band for the **whole curve**, not one point at a time.
- Near the bottom (10th percentile): **−9.8 pp**, band **−24.6 to +5.1**: we cannot tell whether these patients benefit.
- Average: **8.1 pp**, 95% CI **1.4 to 14.9**, which excludes zero.

The estimated average is above zero; the shape of the spread is uncertain.

Why is the band so wide?

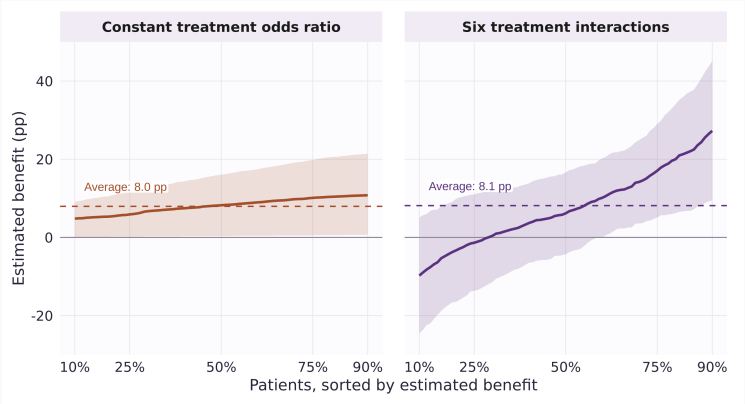
$$\widehat{\mathsf{SPE}}(u) \;\pm\; c \cdot \widehat{\mathsf{se}}(u), \qquad u = 0.10,\, 0.11,\, \ldots,\, 0.90$$

- **Pointwise** 95% interval: $c = 1.96$. Right for one percentile; across 81 percentiles, some would miss.
- **Uniform** 95% band: c covers all 81 at once. It is the 95th percentile of the **largest** standardized deviation across the 81, over 499 bootstrap refits.

At the 10th percentile	c	95% interval (pp)
Pointwise	1.96	−20.2 to +0.6
Uniform, whole curve	2.79	−24.6 to +5.1

Uniform bands as in Chernozhukov, Fernández-Val, and Luo (2018) and SortedEffects; $\widehat{\mathsf{se}}$: bootstrap interquartile range ÷ 1.349. Constant-odds-ratio model: c = 2.17. Band ends rearranged to stay monotone (Chernozhukov, Fernández-Val, and Galichon 2009).

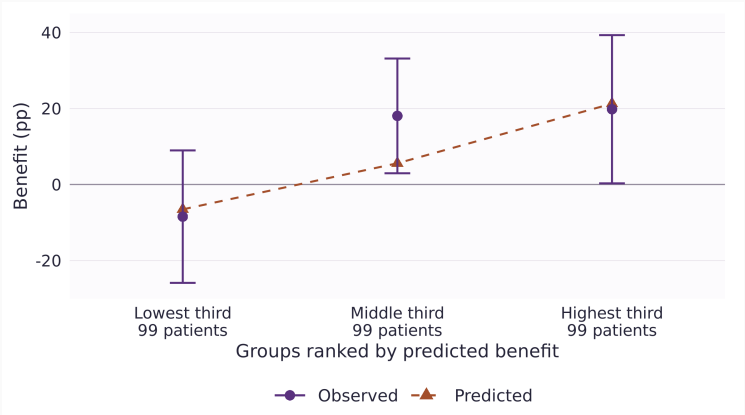
You find what your model allows



- Constant odds ratio: benefit varies only with baseline risk, a narrow spread.
- Six interactions: a much wider spread.
- Same average (**8.0 vs 8.1 pp**); the six interactions improve the fit ($p = .014$).
- More spread is not proof: noise spreads too.

95% bands for the whole curve, one per model. In PATH terms: risk modeling (left) and effect modeling (right) (Kent et al. 2020).

Does the ranking hold up in held-out patients?



- Fit on half the patients (296); rank the other half (297) blind to their outcomes.
- In this one split, higher-ranked patients showed larger observed benefit: slope **1.10** (1 = as predicted, 0 = no signal), adjusted p = .025.
- Lowest third: **−8.5 pp**, CI −25.9 to +9.0: again unclear.

One random split of this trial, balanced by treatment arm: internal validation, not replication. Three groups of 99.

What could fool us?

- **Noise looks like spread:** 19 parameters for 179 deaths; 60 patients per tail.
- **Missing data:** keeping all 618 patients moves the p-value for the interactions from **.014 to .054**.
- **One split, one trial:** colon was the third trial we tried (all three in chapter 04), with no adjustment for that search.

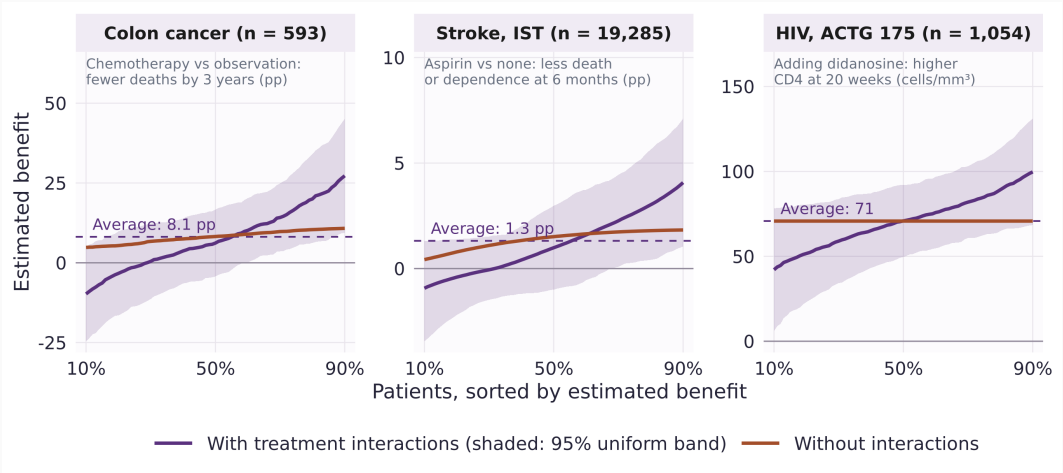
A promising ranking to test in other trials, not a treatment rule.

Discussion

04

What would make sorted effects useful in our own trials?

Three trials, three pictures



Exploratory pilots, plans fixed before fitting; each band within its own model (Moertel et al. 1990; Hammer et al. 1996; International Stroke Trial Collaborative Group 1997). IST data: International Stroke Trial database v2 (Sandercock et al. 2011), doi:10.7488/ds/104, ODC-By 1.0.

Three trials in numbers

	Colon cancer	Stroke (IST)	HIV (ACTG 175)
Patients	593	19,285	1,054
Average benefit (95% CI)	8.1 pp (1.4 to 14.9)	1.3 pp (0.2 to 2.5)	71 cells/mm³ (58 to 84)
10th to 90th percentile	−9.8 to 27.3 pp	−0.9 to 4.1 pp	42 to 100 cells/mm³
Interactions: terms, joint p	6, p = .014	8, p = .35	8, p = .23
Held-out ranking slope (p)	1.10 (p = .025)	0.08 (p = .82)	0.29 (p = .42)

All three interaction curves slope; only colon's held-out check found a ranking signal. Colon was the third trial we looked at.

Interaction models; joint likelihood-ratio tests (colon, IST) or F test (ACTG 175). Held-out slope: 1 = as predicted, 0 = no signal; p for no signal, Holm-adjusted over two models for colon and IST.

What did we learn?

- The average (**8 pp**) hid a wide range of estimated benefits.
- Sorting turned a 19-parameter model into one curve and a profile of who sits at each end.
- The pattern is uncertain and depends on the model; in two other trials, the held-out checks found no ranking signal.

Which result would convince you that the pattern is real?

Next time: causal forests

- **Here:** we chose which interactions to allow; sorted effects then summarized the model's estimates.
- **Causal forests:** let the data choose. A tree splits patients where the effect differs most, say under and over 65; a forest averages many trees.
- **Same caveat:** neither method creates causality. Both need a design, here randomization.

The Effect, §22.3.1 (Huntington-Klein 2025); generalized random forests (Athey, Tibshirani, and Wager 2019). No forest is fitted in this session.

Reading and materials

- **Intuition:** *The Effect*, chapter 10 (treatment effects) and §22.3 (heterogeneous effects), free online (Huntington-Klein 2025).
- **Method and software:** the sorted effects paper and the SortedEffects tutorial (Chernozhukov, Fernández-Val, and Luo 2018; Chen et al. 2020).
- **Risk versus effect modeling:** the PATH statement (Kent et al. 2020).
- **An application (suggested):** blood donors during the COVID-19 pandemic (Bilancini et al. 2022).
- **Code, report and slides:** github.com/valentinklotzbuecher/jcod.

Questions for discussion

- **Spread or noise?** The model estimates a negative benefit for 29% of patient profiles. Would you shrink the interactions first?
- **Planning:** would you prespecify sorted effects in a trial's analysis plan?
- **Risk or interactions:** start from baseline risk, or let the effect depend on traits?
- **Scale:** absolute or relative benefit?

Additional material

+

Model code, the tail-bootstrap failure, full tables and assumptions.

Backup · The model in R

```
1  fm <- outcome ~ treatment + splines::ns(age, df = 3) + female +
2    age_z:female + log_nodes + poor_diff + extensive +
3    obstruct + adhere + surg_late +
4    I((treatment == "chemotherapy") * age_z) +
5    I((treatment == "chemotherapy") * female) +
6    I((treatment == "chemotherapy") * age_z * female) +
7    I((treatment == "chemotherapy") * log_nodes) +
8    I((treatment == "chemotherapy") * poor_diff) +
9    I((treatment == "chemotherapy") * extensive)
10
11 fit <- glm(fm, data = dat, family = binomial())
```

1–3: common model · **4–9:** six treatment interactions · **11:** fit the model.

Comparison model: omit lines 4–9. `age_z = (age - 60) / 10`; `log_nodes = log1p(nodes)`.

Backup · Predict both conditions in R

```r
control <- treated <- dat
control$treatment <- "observation"
treated$treatment <- "chemotherapy"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

- Keep every patient's baseline characteristics fixed.
- Change only treatment assignment.
- Subtract predicted **risks**, not log odds.

Backup · Why the package's tail bootstrap failed here

- In each bootstrap resample, `ca()` refits the logistic model with weights scaled to sum to **20,000**.
- On these data, **141 of 499** refits did not converge, and **137** resamples gave undefined tail means.
- The function still printed normal-looking SEs and p-values. We do not use them.
- Our re-run starts each refit from stable values: **all 499** refits converge, and none are dropped.

Check every bootstrap refit: output that looks normal can still be wrong.

Same resamples, model, tail definitions and tail means; only the numerical starting point changed. Details in the colon report.

Backup · Calculate tail means

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(dat)), c(.10, .90)
)
tails <- list(low = benefit <= cutoff[1],
              high = benefit >= cutoff[2])
traits <- colon_characteristics()$variable
lapply(tails, function(i)
  colMeans(dat[i, traits]))
```

- Select tails by fitted benefit, then summarize baseline characteristics.
- Both tails can contain patients from either randomized arm.

Descriptive calculation only. `R/colon-walkthrough.R` verifies these means against the saved classification table.

Backup · Classification: demographics and nodes

Characteristic	Low	High	Difference	SE	Joint p
Age (years)	43.2	52.9	+9.7	7.1	0.872
Female (%)	100.0	8.3	-91.7	21.4	0.042
Men aged 65+ (%)	0.0	6.7	+6.7	14.2	1.000
Women under 55 (%)	91.7	0.0	-91.7	21.1	0.040
Positive nodes	2.6	8.4	+5.8	2.2	0.297
More than 4 nodes (%)	13.3	88.3	+75.0	22.1	0.132

Difference = high − low. Percentages differ in percentage points.

60 patients per tail. SEs and p-values from our checked re-run of the tail bootstrap (stable starting values, 499 resamples); p-values adjusted across all twelve traits, not just these six.

Backup · Classification: clinical characteristics

Characteristic	Low	High	Difference	SE	Joint p
Poor differentiation (%)	10.0	41.7	+31.7	27.7	0.948
Serosa or beyond (%)	98.3	85.0	-13.3	17.4	0.992
Bowel obstruction (%)	35.0	20.0	-15.0	9.6	0.762
Perforation (%)	3.3	3.3	+0.0	3.7	1.000
Tumour adherence (%)	16.7	15.0	-1.7	10.5	1.000
Long surgery interval (%)	33.3	25.0	-8.3	10.8	0.990

Difference = high − low, in percentage points.

“Long surgery interval”: surgery-to-registration category. SEs and p-values from our checked re-run of the tail bootstrap, adjusted across all twelve traits.

Backup · What is resampled?

- `spe()`**:** refits the model in each resample, then sorts the predicted effects for the original 593 patients.
- `ca()`**:** refits the model in each resample, recomputes who falls in each tail, and recomputes the tail means.
- **Held-out check:** fits on one half only, freezes the ranking, then looks at outcomes in the other half.

Backup · Cohort and sensitivity checks

Analysis	Patients	Average benefit	Interaction p
Complete baseline and endpoint	593	8.1 pp	.014
Missing baseline values with indicators	618	8.4 pp	.054
Early-censored outcome assigned 0	594	8.0 pp	.017
Early-censored outcome assigned 1	594	8.4 pp	.010

- The latter two retain complete baseline variables.
- Eight records disagree on the supplied vs count-derived four-node indicator; the report includes a coding check.

Backup · The other two trial pilots

	Stroke (IST)	HIV (ACTG 175)
Comparison	Aspirin vs none, averaged over heparin	Zidovudine + didanosine vs zidovudine
Outcome	Death or dependence at 6 months	CD4 count at 20 weeks
Models	Logistic: constant odds ratio vs 8 interactions	Linear: main effects vs 8 interactions
Effect may vary with	Age, sex, age × sex, blood pressure, delay, consciousness, atrial fibrillation, leg deficit	Age, sex, weight, CD4, CD8, Karnofsky score, symptoms, prior antiretrovirals
Watch for	Bootstrap ignores hospital clustering; 150 missing outcomes	Public export of two arms; historical therapies

Plans recorded before fitting: `docs/ist-analysis-plan.md`, `docs/actg175-pilot.md`; full IST report on the session page. Native SortedEffects 1.7.0 inference, 499 draws, one held-out split each.

Backup · When is the heterogeneity causal?

- **Identification, both:** randomization or no unmeasured confounding; overlap; consistency/no interference; appropriate handling of selection and missing data.
- **Regression → sorted effects and tail comparison:** reliable conditional-risk estimates; suitable convergence and regularity of the effect distribution.
- **Forests:** enough local data, regularity, suitable nuisance estimates and honest fitting.
- **Inference:** pointwise forest intervals do not automatically cover sorted effects or selected tails.

Both concern conditional average effects. Fitted spread alone does not establish true heterogeneity. (Chernozhukov, Fernández-Val, and Luo 2018; Athey, Tibshirani, and Wager 2019)

Backup · Where do sorted effects fit?

Question	Approach	Watch for
Does the effect differ by one trait?	Subgroup forest plot	One trait at a time; many tests
Does benefit change with baseline risk?	Risk modeling	Misses modifiers unrelated to risk
Which traits modify the effect?	Effect modeling (interactions)	Overfitting; needs validation
How are the estimated effects spread, and who is in the tails?	Sorted effects + classification, on top of a risk or effect model	Only as good as the effect estimates

Sorting summarizes the estimated effects; the model behind them does the estimating. Here: a risk model (constant odds ratio) and an effect model (six interactions).

Risk and effect modeling as defined in the PATH statement (Kent et al. 2020).

Backup · How a forest builds a comparison

For a patient profile, which treated and control patients are comparable?

1 · Grow a tree

Split baseline profiles to find groups with different treatment–control contrasts.

2 · Grow many trees

Different samples and candidate splits give different groups of similar patients.

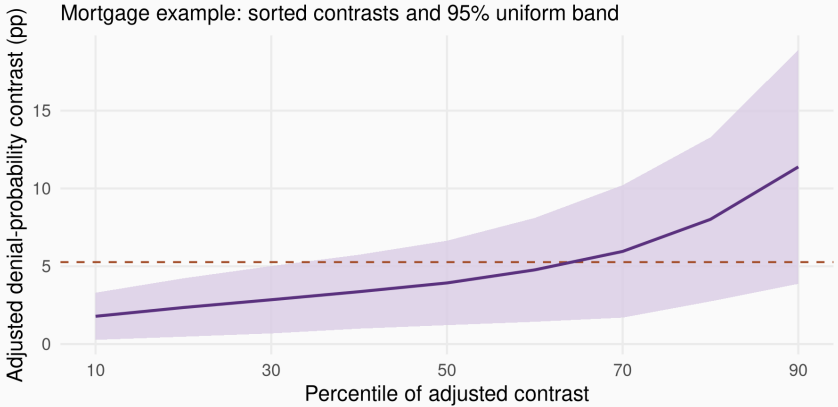
3 · Combine the comparisons

Patients repeatedly grouped with the target profile get more weight in its treatment comparison.

Many trees reduce dependence on any one noisy split.

“Honest” fitting uses separate observations to choose splits and estimate effects. Randomized-trial intuition; no forest was fitted here. (Athey, Tibshirani, and Wager 2019)

Backup · Original published example



Tutorial benchmark: adjusted observational disparity (Munnell et al. 1996; Chen et al. 2020).

References I

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References II

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